SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4303

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool 3 : Case Study

Weightage: 10%

Code of Conduct:

I NISHA K (192411231) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
NISHA K

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

MERIDIAN

Watch demo · How it works · Capabilities · Standards

Register

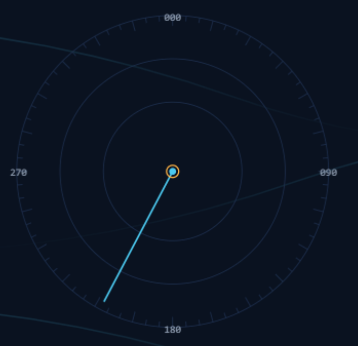
LIVE · SINGLE-STATION DF ENGINE

Locate the transmitter.
Skip the second station.

Meridian fuses Az/El bearing capture with ionospheric modeling — IRI, ASAPS/VOACAP, NeQuick — to project sky-wave HF signals to ground range over a single great-circle path. No baseline. No second site. One antenna, one fix.

Request a demo

See the method

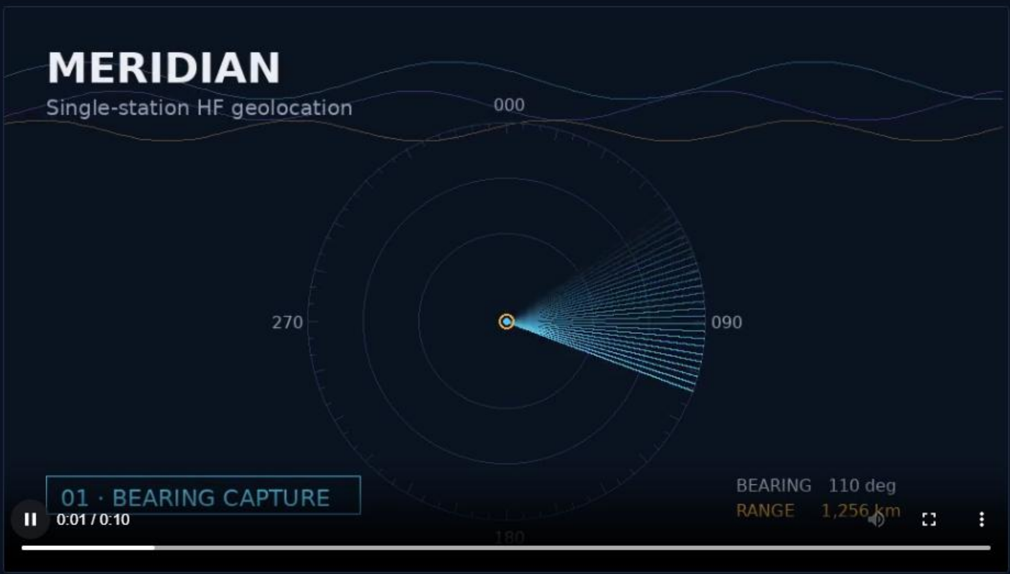


BEARING 208°

RANGE 1,171 km

The full fix, in 10 seconds

Bearing capture, ionospheric modeling, and great-circle projection — watch a single-station fix compute end to end.



THE PROBLEM WITH TRIANGULATION

Two sites, one fix — and everything that can go wrong between them

Multi-station geolocation works, but it demands synchronized clocks, wide baselines, and clean line-of-sight between remote sites. Sky-wave propagation breaks most of those assumptions before you even start.

MULTI-STATION TRIANGULATION	MERIDIAN: SINGLE-STATION FIX
— Requires two or more remote receiver sites	✓ One antenna array, one receiver, one fix
— Needs precise time synchronization between sites	✓ Bearing + ionospheric layer height replace the second site
— Baseline geometry limits usable coverage area	✓ Great-circle projection computes ground range directly
— Comms link between sites is a single point of failure	✓ No inter-site link to maintain or lose
— Deployment cost scales with every added station	✓ Deploys anywhere a single station can reach

THE METHOD

Three measurements. One ground range.

This is the actual computation pipeline, in order — each stage feeds the next.

01 / BEARING

Azimuth & elevation capture

The DF antenna array resolves the incoming sky-wave's azimuth and elevation angle at the receiver.

`Az/El → θ, φ`

→

02 / IONOSPHERE

Layer height modeling

IRI, ASAPS/VOACAP, and NeQuick estimate the reflecting layer's virtual height for the current time, season, and path.

`IRI · ASAPS/VOACAP · NeQuick`

→

03 / PROJECTION

Great-circle ground range

Elevation angle and layer height resolve hop distance; the bearing projects it along a great-circle path to a ground fix.

`θ, φ, h → range, bearing`

CAPABILITIES

Built for the constraints of real HF collection

Everything a field or fixed station needs to turn a single intercept into an actionable fix.

Real-time bearing capture

Continuous Az/El estimation from correlative or Watson-Watt DF antenna arrays.

Live ionospheric modeling

IRI, ASAPS/VOACAP, and NeQuick layer estimates update automatically with time and path geometry.

Great-circle projection

Ground range and fix computed directly from a single intercept — no second site required.

Confidence ellipse output

Every fix ships with an uncertainty ellipse driven by bearing error and layer-height variance.

Multi-hop resolution

Handles 1F and 2F hop ambiguity by cross-checking candidate solutions against the layer model.

Single-site deployment

Runs from one antenna and receiver — field-deployable without a second station or data link.

Register for Meridian

Set up access to request a live demo and technical documentation. Submitted securely over HTTPS via POST.

Full name *

Ada Lovelace

Work email *

ada@organization.com

Organization

Optional

Password *

At least 10 characters

Minimum 10 characters.

Confirm password *

Re-enter password

Create account

Submitted over HTTPS via POST — credentials are never placed in the URL.

ENGINEERING REFERENCE

Web Development Standards Implementation

Five requirements the team hit while building this site, the option chosen for each, and why the alternatives were rejected.

1

Video embedding

Promotional video on the homepage

A ~~`<media src="video.mp4"></media>`~~
Not a real HTML element — browsers won't render it.

✓ `<video src="video.mp4" controls></video>`
Standard HTML5 element built for native video playback.

C ~~`<movie src="video.mp4"></movie>`~~
Invalid tag; not part of the HTML spec.

D ~~`<vid src="video.mp4"></vid>`~~
Invalid tag; likely a typo for <video>.

VERDICT: OPTION B

`<video>` is the only option here that's actually part of the HTML5 spec. It plays media natively with no plugin, and the `controls` attribute gives users play/pause/volume/seek without extra JavaScript.

2

Form submission method

Registration form must securely send user data

A ~~GET~~
Appends data to the URL — logged in browser history and server logs. Never for sensitive data.

✓ POST
Sends data in the request body, not the URL.

C ~~PUT~~
For updating/replacing an existing resource, not creating a new one.

D ~~DELETE~~
Removes a resource — irrelevant to submitting new data.

VERDICT: OPTION B

`POST` keeps credentials and PII out of the URL, browser history, and referrer headers. On its own it doesn't encrypt anything — that protection comes from serving the form over HTTPS (see requirement 5).

3

Semantic navigation element

Navigation menu must follow semantic HTML standards

A ~~<div>~~

Generic container — carries no semantic meaning for browsers or screen readers.

✓ <nav>

Purpose-built landmark element for a block of navigation links.

C ~~<section>~~

A generic thematic grouping — not specific to navigation.

D ~~<header>~~

Introductory content for a page/section; may contain a <nav>, isn't one itself.

VERDICT: OPTION B

`<nav>` lets assistive technology identify and jump to navigation landmarks, and helps search engines parse site structure — meaning conveyed by the tag itself, not just a class name.

4

XML syntax

Configuration files contained syntax errors

A ~~<note><to>User</to><from>Admin</from>~~

Missing closing </note> — the root element is never closed.

✓ <note><to>User</to><from>Admin</from></note>

One root, every tag closed, properly nested.

C ~~<note><to>User</from></note>~~

Mismatched tags — opens <to>, closes </from>.

D ~~<note to="User"></note~~

Closing tag missing its final ">" — incomplete markup.

VERDICT: OPTION B

Well-formed XML needs exactly one root element, every tag closed in the reverse order it opened, quoted attribute values, and escaped reserved characters (`&` , `<` , `>` , `"` , `'`).

5

Web communication protocol

Application must communicate over the correct protocol

A ~~FTP~~

File transfer between systems — not for serving interactive web apps.

✓ `HTTP (as HTTPS in production)`

The protocol for delivering web pages, assets, and form data.

C ~~SMTP~~

For sending email — unrelated to web page delivery.

D ~~SNMP~~

For network device monitoring/management — not web content.

`VERDICT: OPTION B`

HTTP is correct for web delivery — but for anything handling sensitive data, it must be **HTTPS** (HTTP + TLS/SSL), which encrypts the request body in transit. `POST` alone keeps data out of the URL; HTTPS is what stops it being read on the wire.

GET STARTED

See a single-station fix computed live

Bring a signal, or use one of ours. We'll walk through bearing capture, layer modeling, and the resulting ground-range fix in real time.

Request a demo

Read the technical brief